

🚀 JavaScript ES6 Challenge #2 (Bonus)

Scinario

إنت شغال في شركة بتعمل Dashboard لإدارة الموظفين.

عندك الداتا دي:

```
``js
const employees = [
  {
    id: 1,
    name: "Ahmed",
    age: 22,
    salary: 6000,
    department: "IT",
    active: true
  },
  {
    id: 2,
    name: "Sara",
    age: 27,
    salary: 8500,
    department: "HR",
    active: true
  },
  {
    id: 3,
    name: "Ali",
    age: 20,
    salary: 4500,
    department: "IT",
    active: false
  },
  {
    id: 4,
    name: "Mona",
    age: 30,
    salary: 10000,
    department: "Finance",
    active: true
  },
  {

```

```
id: 5,
name: "Omar",
age: 24,
salary: 7000,
department: "Marketing",
active: false
},
{
id: 6,
name: "Youssef",
age: 29,
salary: 12000,
department: "IT",
active: true
}
];
...
---
```

Part 1 - Loops

المطلوب

1

اطبع أسماء كل الموظفين باستخدام

- for
- for...of
- forEach

```
//for
for (let i = 0; i < employees.length; i++) {
    console.log(employees[i].name);
}

//for of
for (let i = 0; i < employees.length; i++) {
    console.log(employees[i].name);
}
```

```

}
//foreach
employees.forEach((employee) => {
  console.log(employee.name);
});

```

2

اطبع الـ Index باستخدام for...in

```

for (let index in employees) {
  console.log(index);
}

```

3

اطبع الموظفين النشطين (active = true) فقط باستخدام Loop عادية.

```

for (let i = 0; i < employees.length; i++) {
  if (employees[i].active === true) {
    console.log(employees[i]);
  }
}

```

Part 2 - ES6

المطلوب

1

حوّل الدالة دي لـ Arrow Function

```

```js

```

```
function welcome(name){
 return "Welcome " + name;
}
...
```

```
const welcome = (name) => `Welcome ${name}`;
console.log(welcome ('Shrouq'))
```

### 2

استخدم Destructuring

```
``js
const employee = employees[0];
...
```

واطلع

- name
- salary

```
const employee = employees[0];
const { name, salary } = employee;
console.log(name);
console.log(salary);
```

### 3

اعمل نسخة جديدة من الـ employee باستخدام Spread Operator وزود فيها

```
``js
country:"Egypt"
...
```

---

### 4

استخدم Template Literal واطبع

...

Ahmed works in IT and earns 6000

...

```
console.log(`${employees[0].name} works in ${employees[0].department} and
earns ${employees[0].salary}`);
```

# Part 3 - map()

### 1

اعمل Array فيها أسماء الموظفين بس.

```
const employeeNames = employees.map((employee) => employee.name);

console.log(employeeNames);
```

### 2

اعمل Array فيها الرواتب فقط.

```
const salaries = employees.map((employee) => employee.salary);

console.log(salaries);
```

### 3

اعمل Array بالشكل ده

```
``js
[
 "Ahmed (IT)",
 "Sara (HR)"
]
``
```

```
const nameAndDepartment = employees.map(
 (employee) => `${employee.name} (${employee.department})`
);

console.log(nameAndDepartment);
```

### 4

اعمل Array جديدة تزود مرتب كل موظف 1000 جنيه.

< بدون تعديل الـ Array الأصلية.

```
const increasedSalaries = employees.map((employee) => {
 return {
 ...employee,
 salary: employee.salary + 1000
 };
});

console.log(increasedSalaries);
```

## # Part 4 - filter()

#### طلع

1. الموظفين اللي مرتباتهم أكبر من 7000.

```
const highSalary = employees.filter(
 employee => employee.salary > 7000
);
console.log(highSalary);
```

2. الموظفين اللي في قسم IT.

```
const itEmployees = employees.filter(
 (employee) => employee.department === "IT"
);
console.log(itEmployees);
```

3. الموظفين النشطين.

```
const activeEmployees = employees.filter(
 employee => employee.active === true
);
console.log(activeEmployees);
```

4. الموظفين سنهم أقل من 25.

```
const youngEmployees = employees.filter(
 (employee) => employee.age < 25
);
console.log(youngEmployees)
```

5. الموظفين اللي في قسم IT ومرتبهم أكبر من 5000.

```
const itHighSalary = employees.filter(
 (employee) =>
 employee.department === "IT" &&
 employee.salary > 5000
);
console.log(itHighSalary)
```

## # Part 5 - find()

#### هات

1. أول موظف مرتبه أكبر من 9000.

```
const firstHighSalary = employees.find(
 (employee) => employee.salary > 9000
);
console.log(firstHighSalary)
```

2. أول موظف في قسم HR.

```
const firstHR = employees.find(
 (employee) => employee.department === "HR"
);
```



```
(console.log(firstHR
```

3. أول موظف غير نشط.

```
const firstInactive = employees.find(
 employee => employee.active === false
);
(console.log(firstInactive
```

4. جَرِّبْ تَجِيبْ موظف الـ id بتاعه 100.

إيه اللي هيحصل؟

```
const employee100 = employees.find(
 (employee) => employee.id === 100
);
console.log(employee100) //undefined because there is no id "100"
```

## # Part 6 - Mixed Challenge

## المطلوب

اعمل النتائج دي بدون استخدام Loop عادية.

---

### 1

اعمل Array فيها أسماء الموظفين النشطين فقط.

```
const activeNames = employees.filter((employee) => employee.active)
 .map((employee) => employee.name);
console.log(activeNames)
```

### 2

اعمل Array فيها أسماء موظفين الـ IT فقط.

```
const itNames = employees
 .filter((employee) => employee.department === "IT")
 .map((employee) => employee.name);
console.log(itNames)
```

### 3

اعمل Array فيها أسماء الموظفين اللي مرتبهم أكبر من 7000.

```
const highSalaryNames = employees
 .filter((employee) => employee.salary > 7000)
 .map((employee) => employee.name);
 console.log(highSalaryNames)
```

### 4

اعمل Array بالشكل ده

```
``js
[
{
 employee:"Ahmed",
 bonus:600
},
...
]
```

< اليونس = 10% من المرتب.

```
const bonuses = employees.map((employee) => {
 return {
 employee: employee.name,
 bonus: employee.salary * 0.1
 };
});

console.log(bonuses);
```

### 5

اعمل Array فيها أول حرف من اسم كل موظف.

مثال

```
``js
["A","S","A","M"]
```

...

```
const firstLetters = employees.map(
 (employee) => employee.name[0]
);

console.log(firstLetters);
```

## # Part 7 - Logical Thninking

#### عندك

```
``js
const numbers = [5,12,8,20,15,30,3,40];
...
```

#### المطلوب

1. طلع الأرقام الأكبر من 10.

```
const numbers = [5, 12, 8, 20, 15, 30, 3, 40];
const greaterThanTen = numbers.filter(
 number => number > 10
);

console.log(greaterThanTen);
```

2. اضرب كل رقم  $\times 2$ .

```
const numbers = [5, 12, 8, 20, 15, 30, 3, 40];
const doubledNumbers = numbers.map(
 (number) => number * 2
);
console.log(doubledNumbers);
```

3. هات أول رقم أكبر من 25.

```
;[const numbers = [5, 12, 8, 20, 15, 30, 3, 40
)const firstGreaterThan25 = numbers.find
 number) => number > 25)
; (
; (console.log(firstGreaterThan25
```

4. اطبع كل الأرقام.

```
;[const numbers = [5, 12, 8, 20, 15, 30, 3, 40
} <= (numbers.forEach((number
; (console.log(number
; ({
```

5. اعمل Array فيها String بالشكل

...

Number is 5

Number is 12

...

```
const numbers = [5, 12, 8, 20, 15, 30, 3, 40];
const numberStrings = numbers.map(
 (number) => `Number is ${number}`
);
console.log(numberStrings);
```

## # Part 8 - Object Challenge

عندك

```
```js
const product = {
  id:1,
  title:"Laptop",
  price:25000,
  category:"Electronics"
}
...
---
```

المطلوب

1. اطبع كل الـ Keys باستخدام `.for...in`.

```
const product = {
  id: 1,
  title: "Laptop",
  price: 25000,
  category: "Electronics"
};

for (let key in product) {
  console.log(key);
}
```

```
}
```

2

اطبع كل الـ Values.

```
const product = {
  id: 1,
  title: "Laptop",
  price: 25000,
  category: "Electronics"
};

for (let key in product) {
  console.log(product[key]);
}
```

3

اعمل Object جديد وضيف

```
``js
stock:15
``
```

```
const product = {
  id: 1,
  title: "Laptop",
  price: 25000,
  category: "Electronics"
};
```

```
const newProduct = { ...product, stock: 15};  
console.log(newProduct);
```

4

استخدم Destructuring.

```
const product = {  
  id: 1,  
  title: "Laptop",  
  price: 25000,  
  category: "Electronics"  
};  
  
const { id, title, price, category } = product;  
  
console.log(id);  
console.log(title);  
console.log(price);  
console.log(category);
```

Part 9 - Mini Dashboard

اعمل Function اسمها

```
``js  
dashboard()  
``
```

لما تتنادى تطبع بالشكل ده

...

Total Employees : 6

Active Employees : 5

Inactive Employees : 1

IT Employees : 1

Highest Salary : 5000

First HR Employee : 1

Employee Names :

Ahmed

Sara

Ali

...

...

```
function dashboard() {

    const activeEmployees = employees.filter(
        (employee) => employee.active
    );

    const inactiveEmployees = employees.filter(
        (employee) => !employee.active
    );

    const itEmployees = employees.filter(
        (employee) => employee.department === "IT"
    );

    const highestSalary = Math.max(
```

```
        ...employees.map((employee) => employee.salary)
    );

    const firstHR = employees.find(
        (employee) => employee.department === "HR"
    );

    const names = employees.map(
        (employee) => employee.name
    );

    console.log(`Total Employees : ${employees.length}`);

    console.log(`Active Employees : ${activeEmployees.length}`);

    console.log(`Inactive Employees : ${inactiveEmployees.length}`);

    console.log(`IT Employees : ${itEmployees.length}`);

    console.log(`Highest Salary : ${highestSalary}`);

    console.log(`First HR Employee : ${firstHR.name}`);

    console.log("Employee Names :");

    names.forEach((name) => {
        console.log(name);
    });
}

dashboard();
```

Bonus

لو تعرف `reduce`

بدون Loop عادية احسب

1

إجمالي المرتبات.

```
const totalSalaries = employees.reduce(
  (total, employee) => total + employee.salary,
  0
);
console.log(totalSalaries);
```

2

متوسط المرتبات.

```
const averageSalary = totalSalaries / employees.length;

console.log(averageSalary);
```

3

أعلى مرتب.



```
const highestSalaryReduce = employees.reduce(  
  (highest, employee) => {  
    return employee.salary > highest  
      ? employee.salary  
      : highest;  
  },  
  0  
);  
  
console.log(highestSalaryReduce);
```

4

عدد الموظفين النشطين.

```
const activeCount = employees.reduce(  
  (count, employee) => {  
    return employee.active ? count + 1 : count;  
  },  
  0  
);  
  
console.log(activeCount);
```